

SECTION B

9 (**long**) The bilinear transform is used in digital signal processing and maps the variable z to the variable s as

$$s = 2 \left(\frac{z - 1}{z + 1} \right)$$

where both s and z are complex variables. For example, the point $z = 0$ is “transformed” to the point $s = -2$ in the complex plane.

(a) Show that the inverse transform expressing z in terms of s is $z = \frac{1+s/2}{1-s/2}$. [3]

(b) By expressing s as $s = ix$ for a real variable x , show that z is on the unit circle if s is on the imaginary axis. [8]

(c) Now expressing z as $z = e^{i\theta}$ for an angle θ , give an expression for the corresponding s and confirm that it is on the imaginary axis. [8]

(d) Find the doubling points of the bilinear transform, i.e., the points z for which $s = 2z$. [5]

(e) The inverse bilinear transform is an approximation for the exponential transform $z = e^s$. Show that this approximation is accurate up to the quadratic term. [6]